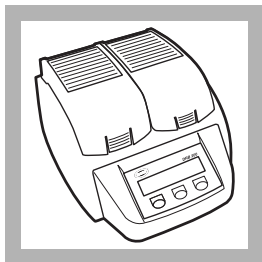


PHOSPHORUS, TOTAL, HR (0.0 to 100.0 mg/L PO₄³⁻)

Molybdovanadate Method with Acid Persulfate Digestion*

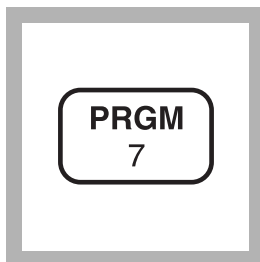
Test 'N Tube™ Procedure

For water and wastewater



1. Turn on the DRB200 Reactor. Heat to 150 °C.

Note: See DRB200 instrument manual for selecting preprogrammed temperature applications

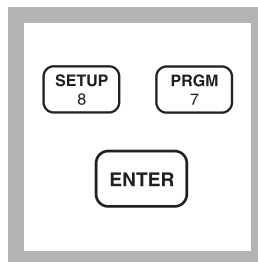


2. Enter the stored program number for phosphorus total high range, Test 'N Tube.

Press: **PRGM**

The display will show:

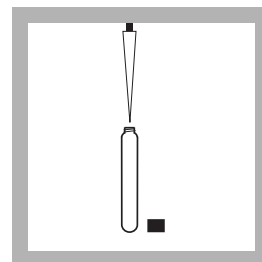
PRGM ?



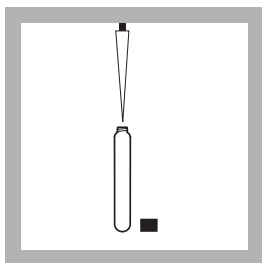
3. Press: **87 ENTER**

The display will show **mg/L, PO₄** and the **ZERO** icon.

Note: For alternate forms (P, P₂O₅), press the **CONC** key.

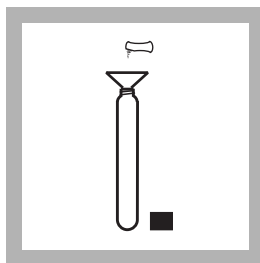


4. Use a TenSette® Pipet to add 5.0 mL of deionized water to a Total Phosphorus Test 'N Tube Vial (the blank).

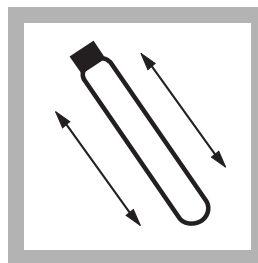


5. Use a TenSette Pipet to add 5.0 mL of sample to a Total Phosphorus Test 'N Tube Vial (the sample).

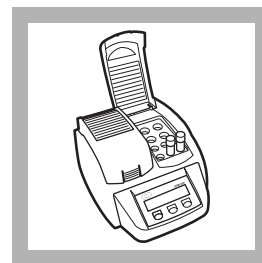
Note: Adjust the pH of stored samples to 6–8 before analysis.



6. Use a funnel to add the contents of one Potassium Persulfate Powder Pillow for Phosphonate to each vial.



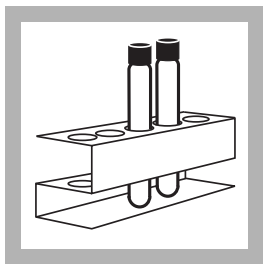
7. Cap tightly and shake to dissolve.



8. Place the vials in the DRB200 Reactor. Heat for 30 minutes.

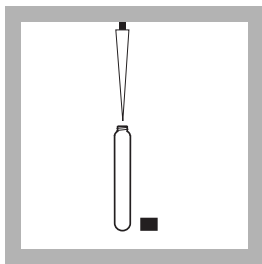
Press: **TIMER ENTER** to time the heating period.

* Adapted from *Standard Methods for the Examination of Water and Wastewater*.

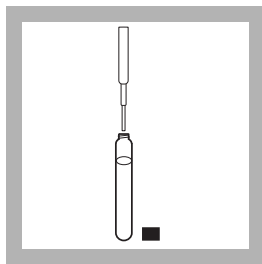


9. Carefully remove the vials from the reactor. Place them in a test tube rack and allow to cool to room temperature (18–25 °C).

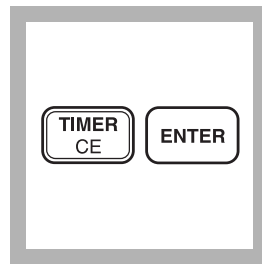
Note: Vials will be hot.



10. Use a TenSette Pipet to add 2.0 mL of 1.54 N sodium hydroxide to each vial. Cap and invert to mix.



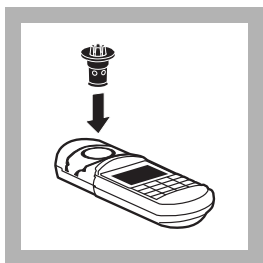
11. Use a polyethylene dropper to add 0.5 mL of Molybdovanadate Reagent to each vial. Cap and invert to mix.



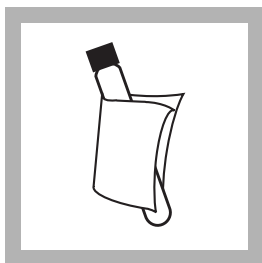
12. Press:
TIMER ENTER

A 7-minute reaction period will begin.

Note: Read the samples between 7 and 9 minutes.

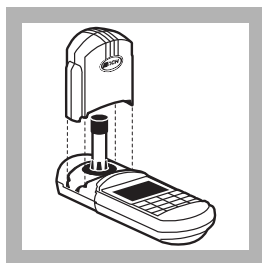


13. Insert the COD/TNT Adapter into the cell holder by rotating the adapter until it drops into place. Then push down to fully insert it.



14. Clean the outside of the vials with a towel.

Note: Wipe with a damp towel, followed by a dry one, to remove fingerprints or other marks.

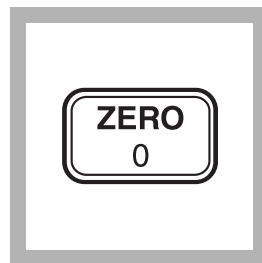


15. When the timer sounds, place the blank vial in the adapter.

Push straight down on the top of the vial until it seats solidly into the adapter.

Tightly cover the vial with the instrument cap.

Note: Do not move the vial from side to side as this can cause errors.

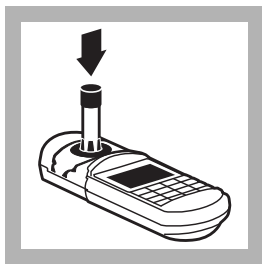


16. Press: **ZERO**

The cursor will move to the right, then the display will show:

0.0 mg/L PO₄

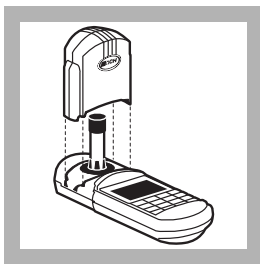
Note: Reagent blanks for each lot of reagents may be used more than once, but should not be used for longer than one day.



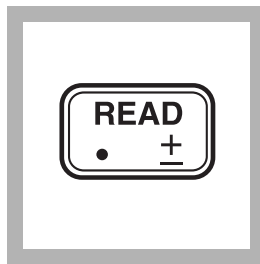
17. Place the prepared sample vial in the adapter.

Push straight down on the top of the vial until it seats solidly into the adapter.

Note: Do not move the vial from side to side as this can cause errors.



18. Tightly cover the vial with the instrument cap.



19. Press: **READ**

The cursor will move to the right, then the result in mg/L phosphate (PO_4^{3-}) will be displayed.

Note: For best results, use Standard Adjust with each new lot of reagent. (See Accuracy Check.)

Sampling and Storage

Collect samples in plastic or glass bottles that have been acid cleaned with 1:1 Hydrochloric Acid Solution and rinsed with deionized water. Do not use commercial detergents containing phosphates for cleaning the glassware used in this test.

Analyze samples immediately after collection for best results. If prompt analysis is impossible, preserve the sample for up to 28 days by adjusting the pH to 2 or less with concentrated H_2SO_4 (about 2 mL per liter) and storing at 4 °C. Warm the sample to room temperature and neutralize with 5.0 N NaOH before analysis.

Correct test results for volume additions; see *Volume Additions* in Section 1 of the *DR/890 Procedures Manual*.

Accuracy Check

Note: Clean glassware with 1:1 hydrochloric acid solution. Rinse again with deionized water. Do not use detergents containing phosphates to clean glassware.

Standard Additions Method

- a. Fill each of three 10-mL graduated mixing cylinders with 10 mL of sample.
- b. Snap the neck off a 10-mL Voluette® Ampule of Phosphate Standard Solution, 500 mg/L as PO_4^{3-} (Cat. No. 14242-10).
- c. Use a TenSette Pipet to add 0.1 mL, 0.2 mL and 0.3 mL, respectively, to the three 10-mL aliquots of the water sample prepared in *step a*. Mix well.
- d. Analyze samples from *step c* as described in the procedure. Use 5.0 mL of the prepared sample for each test. The concentration should increase: 5 mg/L, 10 mg/L, and 15 mg/L PO_4^{3-} , respectively.
- e. If these increases do not occur, see *Standard Additions (Section 1 of the DR/890 Procedures Manual)* for more information.

Standard Solution Method

To check accuracy, prepare an 80 mg/L standard by pipetting 8.0 mL of solution from a 10-mL Voluette® Ampule of Phosphate Standard Solution, 500 mg/L as PO_4^{3-} into an acid-cleaned, Class A, 50-mL volumetric flask. Dilute to the mark with deionized water. Substitute this standard for the sample and perform the procedure as described.

Standard Adjust

To adjust the calibration curve using the reading obtained with the 80 mg/L PO_4^{3-} standard solution, press the **SETUP** key and scroll, using the arrow keys, to the **STO** option. Press **ENTER** to activate the standard adjust option. Then enter 80.0 to edit the standard concentration to match that of the standard used. Press **ENTER** to complete the adjustment. See *Standard Curve Adjustment, Section 1 of the Procedures Manual* for more information.

Interferences

Large amounts of sample turbidity may cause inconsistent results in the test because the acid present in the reagents may dissolve some of the suspended particles and because of variable desorption of orthophosphate from the particles.

The following may interfere when present in concentrations exceeding those listed below:

Interfering Substance	Interference Level and Treatment
Arsenate	Causes positive interference if the sample is heated. ¹
Iron, ferrous	Blue color caused by ferrous iron does not interfere if iron concentration is less than 100 mg/L.
Molybdate	Causes negative interference above 1000 mg/L.
Silica	Causes positive interference if the sample is heated.*
Extreme pH or highly buffered samples	May exceed buffering capacity of the reagents. See <i>pH Interferences</i> in <i>Section 1</i> of the <i>DR/890 Procedures Manual</i> . Samples may require pretreatment. Sample pH should be about 7.
Fluoride, thorium, bismuth, thiosulfate or thiocyanate	Cause a negative interference.
Temperature, Cold (less than 18 °C)	Causes a negative interference.
Temperature, Hot (greater than 25 °C)	Causes a positive interference. Post-digestion samples should be brought to room temperature (18–25 °C) before the addition of the Molybdovanadate Reagent or sodium hydroxide.
The following do not interfere in concentrations up to 1000 mg/L: Pyrophosphate, tetraborate, selenate, benzoate, citrate, oxalate, lactate, tartrate, formate, salicylate, Al ³⁺ , Fe ³⁺ , Mg ²⁺ , Ca ²⁺ , Ba ²⁺ , Sr ²⁺ , Li ⁺ , Na ⁺ , K ⁺ , NH ₄ ⁺ , Cd ²⁺ , Mn ²⁺ , NO ₃ ⁻ , NO ₂ ⁻ , SO ₄ ²⁻ , SO ₃ ²⁻ , Pb ²⁺ , Hg ⁺ , Hg ²⁺ , Sn ²⁺ , Cu ²⁺ , Ni ²⁺ , Ag ⁺ , U ⁴⁺ , Zr ⁴⁺ , AsO ₃ ⁻ , Br ⁻ , CO ₃ ²⁻ , ClO ₄ ⁻ , CN ⁻ , IO ₃ ⁻ , SiO ₄ ⁴⁻ .	

¹ Gentle warming of the sample to reach room temperature will not cause this substance to interfere.

Method Performance

Precision

In a single laboratory, using a standard solution of 80.0 mg/L PO_4^{3-} and two lots of reagent with the instrument, a single operator obtained a standard deviation of ± 3.0 mg/L PO_4^{3-} .

Estimated Detection Limit

The estimated detection limit for program 87 is 7.0 mg/L PO_4^{3-} . For more information on the estimated detection limit, see *Section 1* of the *DR/890 Procedures Manual*.

Safety

Good safety habits and laboratory techniques should be used throughout the procedure. Consult the Material Safety Data Sheet for information specific to the reagents used.

Sample Disposal Information

The final samples will contain molybdenum. In addition, the final samples will have a pH less than 2 and are considered corrosive (D002) by the Federal RCRA. Consult the Material Safety Data Sheet for information specific to the reagent used.

Summary of Method

Phosphates present in organic and condensed inorganic forms (meta-, pyro- or other polyphosphates) must be converted to reactive orthophosphate before analysis. Pretreatment of the sample with acid and heat provides the conditions for hydrolysis of the condensed inorganic forms. Organic phosphates are converted to orthophosphate by heating with acid and persulfate.

Orthophosphate reacts with molybdate in an acid medium to produce a phosphomolybdate complex. In the presence of vanadium, yellow vanadomolybdophosphoric acid forms. The intensity of the yellow color is proportional to the phosphate concentration.

Installing this Program on the DR/800

This procedure will add the current method as a new Hach program to your DR/800.

1. Turn the DR/800 on by pressing the **ON** key.
2. Press the **SETUP** key.
3. Press the down arrow key two times so that the prompt line shows **USER**.
4. Press the **ENTER** key.
5. Enter **8138**, followed by **ENTER**.
6. Enter each of the numbers in the right column, each followed by **ENTER**. The line numbers in the left column relate to the line number on the display. At any time you may use the arrow keys to scroll back to review or change any number you have already entered.

Line Number	Entry	Line Number	Entry
1	87	18	0
2	4	19	0
3	73	20	80
4	0	21	79
5	0	22	52
6	0	23	0
7	0	24	0
8	0	25	80
9	0	26	0
10	0	27	0
11	0	28	0
12	66	29	0
13	175	30	80
14	48	31	50
15	32	32	79
16	0	33	53
17	0	34	0
35	62	46	0

PHOSPHORUS, TOTAL, HR, continued

Line Number	Entry	Line Number	Entry
36	166	47	15
37	246	48	7
38	148	49	8
39	63	50	1
40	63	51	164
41	78	52	0
42	252	53	0
43	4	54	40
44	76	55	0
45	128	56	255

REQUIRED REAGENTS

Total High Range Phosphorus Test 'N Tube™ Reagent Set 50 vials 27672-45
Includes: (50) Total Phosphorus Test 'N Tube™ Vials*, (2) 272-42, (1) 20847-66
(1) 20760-26, (1) 27430-42

Description	Quantity Required		Cat. No.
	Per Test	Unit	
Molybdovanadate Reagent	0.5 mL .. 25 mL		20760-26
Potassium Persulfate Powder Pillows.....	1	50/pkg	20847-66
Sodium Hydroxide Solution, 1.54 N	2 mL .. 100 mL		27430-42
Total Phosphorus Test 'N Tube™ Vials.....	1	50/pkg	*
Water, deionized.....	100 mL		272-42

REQUIRED APPARATUS

DRB 200 Reactor, 110 V, 15 x 16 mm tubes LTV082.53.40001
DRB 200 Reactor, 220 V, 15 x 16 mm tubes..... LTV082.52.40001
COD/TNT Adapter, DR/800 series..... 1 each | | 48464-00 || Dropper, LDPE, 0.5 to 1.0 mL..... | 1 | 20/pkg | 21247-20 |
Pipet, TenSette®, 1 to 10 mL	1	each	19700-10
Pipet Tips, for 19700-10 TenSette® Pipet.....	varies ..	50/pkg	21997-96
Test Tube Rack	1-3	each	18641-00

* These items are not sold separately.

PHOSPHORUS, TOTAL, HR, continued

OPTIONAL REAGENTS

Description	Unit	Cat. No.
Hydrochloric Acid Standard Solution, 6.0 N (1:1)	500 mL	884-49
Phosphate Standard Solution, PourRite™ ampule, 500 mg/L as PO ₄ ³⁻ , 2-mL.....	20/pkg.....	14242-20
Phosphate Standard Solution, Voluette™ ampule, 500 mg/L as PO ₄ ³⁻ , 10-mL.....	16/pkg.....	14242-10
Sodium Hydroxide Standard Solution, 5.0 N.....	1 L	2450-53
Sulfuric Acid, ACS, concentrated	500 mL	979-491
Wastewater Influent Standard, Inorganic (NH ₃ -N, NO ₃ -N, PO ₄ , COD, SO ₄ , TOC)	500 mL	28331-49

OPTIONAL APPARATUS

Ampule Breaker Kit	each.....	21968-00
Aspirator, vacuum	each.....	2131-00
Cylinder, graduated, mixing, 10 mL (3 required)	each.....	20886-38
DRB 200 Reactor, 110 V, 21 x 16 mm and 4 x 20 mm.....	LTV082.53.42001	
DRB 200 Reactor, 220 V, 21 x 16 mm and 4 x 20 mm.....	LTV082.52.42001	
DRB 200 Reactor, 110 V, 9 x 16 mm and 2 x 20 mm.....	LTV082.53.30001	
DRB 200 Reactor, 220 V, 9 x 16 mm and 2 x 20 mm.....	LTV082.52.30001	
Filter Holder, 47 mm, 300 mL, graduated.....	each.....	13529-00
Filter, membrane, 47 mm, 0.45 microns	200/pkg.....	13530-01
Flask, filtering, 500-mL	each.....	546-49
Flask, volumetric, Class A, 50-mL	each.....	14574-41
pH Indicator Paper, 1 to 11 pH units.....	5 rolls/pkg.....	391-33
pH Meter, <i>sens^{ion}</i> ™1, portable with electrode	each.....	51700-10
Pipet Filler, Safety Bulb	each.....	14651-00
Pipet, TenSette®, 0.- to 1.0-mL	each.....	19700-01
Pipet Tips, for 19700-01.....	50/pkg.....	21856-96
Pipet Tips, for 19700-01	1000/pkg.....	21856-28
Pipet, volumetric, Class A, 8.00-mL	each.....	14515-08
Stopper, No. 7 one hole	6/pkg.....	2119-07
Tubing, rubber	12 feet.....	560-19